MedCore Clinical EHR & Patient Portal API Reference

Version 4.1 of the MedCore Clinical API enables hospitals, outpatient clinics and telehealth platforms to integrate with patient charts, scheduling, e-prescribing, diagnostic laboratories and insurance claim adjudication.

Base URL

All clinical API calls are made to `https://api.medcorehealth.org/v4`.

Authentication

All requests must include a SMART-on-FHIR Bearer token in the `Authorization` header: `Authorization: Bearer <access_token>`.

Patient Charts & Demographics

Search patient records

`GET /patients`

Searches clinical patient charts by medical record number (MRN), family name or date of birth within the authenticated facility.

Parameters:

Parameter	In	Type	Required	Description
limit	query	integer	No	Maximum number of clinical records to return (1-100). Defaults to 25.
mrn	query	string	No	Exact hospital medical record number (MRN).
date_of_birth	query	string	No	Patient date of birth in ISO 8601 YYYY-MM-DD format.
X-Facility-ID	query	string	No	Identifier of the hospital or outpatient clinic facility performing the lookup.

Response 200 OK:

```
{
  "data": [
    {
      "id": "pat_10293847",
      "patient_mrn": "MRN-884920",
      "given_name": "Hannah",
      "family_name": "Abbott",
      "date_of_birth": "1984-07-19",
      "biological_sex": "female",
      "phone": "+16175550192",
      "email": "hannah.abbott@example.org",
      "allergies": ["Penicillin"],
      "insurance_policy": {
        "payer_name": "BlueShield National",
        "member_id": "BSN-9920182",
        "group_number": "GRP-4410"
      },
      "created_at": "2025-11-04T09:15:00Z"
    }
  ],
  "total_count": 1
}
```

Returns `401 Unauthorized` when the bearer token is invalid.

Register a new patient

`POST /patients`

Creates a new master patient index (MPI) chart and assigns a permanent medical record number.

Request body:

```
{
  "given_name": "Hannah",
  "family_name": "Abbott",
  "date_of_birth": "1984-07-19",
  "biological_sex": "female",
  "phone": "+16175550192",
  "email": "hannah.abbott@example.org",
  "insurance_policy": {
    "payer_name": "BlueShield National",
    "member_id": "BSN-9920182"
  }
}
```

Response 201 Created:

```
{
  "id": "pat_10293847",
  "patient_mrn": "MRN-884920",
  "given_name": "Hannah",
  "family_name": "Abbott",
  "date_of_birth": "1984-07-19",
  "biological_sex": "female",
  "insurance_policy": {
    "payer_name": "BlueShield National",
    "member_id": "BSN-9920182",
    "group_number": "GRP-4410"
  },
  "created_at": "2025-11-04T09:15:00Z"
}
```

Error codes:

Status	Description
400	Invalid clinical payload or missing mandatory field.
409	A patient chart with the same MRN or national health identifier already exists.

Retrieve patient chart

`GET /patients/{patient_id}`

Retrieves full patient demographics, active allergies, primary care provider and insurance coverage.

Path parameters:

Parameter	In	Type	Required	Description
patient_id	path	integer	Yes	Unique identifier of the patient chart.

Response 200 OK:

```
{
  "id": "pat_10293847",
  "patient_mrn": "MRN-884920",
  "given_name": "Hannah",
  "family_name": "Abbott",
  "date_of_birth": "1984-07-19",
  "biological_sex": "female",
  "insurance_policy": {
    "payer_name": "BlueShield National",
    "member_id": "BSN-9920182",
    "group_number": "GRP-4410"
  },
  "created_at": "2025-11-04T09:15:00Z"
}
```

Returns `404 Not Found` if the patient chart does not exist.

Update patient demographics

`PATCH /patients/{patient_id}`

Updates contact phone, residential address or primary insurance policy on an existing patient chart.

Path parameters:

Parameter	In	Type	Required	Description
patient_id	path	string	Yes	Unique UUID identifier of the patient chart.

Request body:

Field	Type	Required	Description
phone	string	No	Updated phone number.
email	string	No	Updated email address.

Response 200 OK:

```
{
  "id": "pat_10293847",
  "patient_mrn": "MRN-884920",
  "given_name": "Hannah",
  "family_name": "Abbott",
  "date_of_birth": "1984-07-19",
  "biological_sex": "female",
  "insurance_policy": {
    "payer_name": "BlueShield National",
    "member_id": "BSN-9920182",
    "group_number": "GRP-4410"
  },
  "created_at": "2025-11-04T09:15:00Z"
}
```

Get patient billing summary

`GET /patients/{patient_id}/billing-summary`

Returns the patient's outstanding self-pay balance and payment plan schedule.

```
{
  "outstanding_balance_cents": 14500,
  "last_statement_date": "2026-01-15"
}
```

Outpatient Appointments

List clinical appointments

`GET /appointments`

Returns scheduled, checked-in and completed outpatient encounters filtered by clinician, department or date.

Query parameters:

Parameter	In	Type	Required	Description
limit	query	integer	No	Maximum number of clinical records to return.
provider_id	query	integer	No	Filter appointments for a specific attending clinician.
clinic_date	query	string	No	Filter appointments scheduled on this calendar date (YYYY-MM-DD).

Response 200 OK:

```
{
  "data": [
    {
      "id": "apt_330192",
      "patient_id": "pat_10293847",
      "provider_id": "prv_88210",
      "department": "Cardiology",
      "start_time": "2026-03-10T14:00:00Z",
      "duration_minutes": 30,
      "visit_modality": "in_person",
```

```
      "status": "scheduled"
    }
  ],
  "total_count": 1
}
```

Book an appointment

`POST /appointments`

Books an outpatient or telehealth appointment slot between a patient and an attending clinician.

Request body:

Field	Type	Required	Description
patient_id	string	Yes	Patient UUID.
provider_id	string	Yes	Attending clinician UUID.
department	string	Yes	Clinical department name.
start_time	string	Yes	ISO 8601 appointment start timestamp.
duration_minutes	integer	Yes	Length of the appointment in minutes.
visitModality	string	Yes	Modality (`in_person` or `telehealth`).
chief_complaint	string	Yes	Reason for visit.

Response 201 Created:

```
{
  "id": "apt_330192",
  "patient_id": "pat_10293847",
  "provider_id": "prv_88210",
  "department": "Cardiology",
  "start_time": "2026-03-10T14:00:00Z",
  "duration_minutes": 30,
  "visit_modality": "in_person",
  "status": "scheduled"
}
```

Returns `409 Conflict` if the clinician is already booked.

Reschedule an appointment

`PATCH /appointments/{appointment_id}/reschedule`

Transfers the patient's encounter to a different specialist department and generates a new specialist referral authorization request.

Path parameters:

Parameter	In	Type	Required	Description
appointment_id	path	string	Yes	Unique UUID identifier of the clinical appointment.

Request body:

Field	Type	Required	Description
new_start_time	string	Yes	New start timestamp.
reason	string	Yes	Reason for rescheduling.

Response 200 OK:

```
{
  "id": "apt_330192",
  "patient_id": "pat_10293847",
  "provider_id": "prv_88210",
  "department": "Cardiology",
  "start_time": "2026-03-12T10:30:00Z",
  "duration_minutes": 30,
```

```
  "visit_modality": "in_person",
  "status": "scheduled"
}
```

Prescriptions (eRx)

List active prescriptions

`GET /prescriptions`

Returns active and historical medication orders for a given patient.

Query parameters:

Parameter	In	Type	Required	Description
patient_id	query	string	Yes	Unique UUID identifier of the patient.
active_only	query	boolean	No	When true, excludes expired and discontinued medication orders.

Response 200 OK:

```
{
  "data": [
    {
      "id": "rx_661029",
      "patient_id": "pat_10293847",
      "prescriber_id": "prv_88210",
      "rxnorm_code": "860975",
      "medication_name": "Metformin Hydrochloride",
      "dosage_mg": 500,
      "refills_remaining": 3,
      "status": "active",
      "dosage_instructions": {
        "route": "oral",
        "frequency_per_day": 2,
        "duration_days": 90,
        "take_with_food": true
      },
      "issued_at": "2026-01-12T11:20:00Z"
    }
  ]
}
```

Issue an e-prescription

`POST /prescriptions`

Transmits an electronic medication prescription (eRx) to the patient's designated retail or mail-order pharmacy after checking drug-allergy interactions.

Request body:

```
{
  "patient_id": "pat_10293847",
  "prescriber_id": "prv_88210",
  "rxnorm_code": "860975",
  "medication_name": "Metformin Hydrochloride",
  "refills_allowed": 3,
  "dosage_instructions": {
    "route": "oral",
    "frequency_per_day": 2,
    "duration_days": 90
  }
}
```

Response 201 Created:

```
{
  "id": "rx_661029",
  "patient_id": "pat_10293847",
  "prescriber_id": "prv_88210",
  "rxnorm_code": "860975",
```

```
  "medication_name": "Metformin Hydrochloride",
  "dosage_mg": 500,
  "refills_remaining": 3,
  "status": "active",
  "dosage_instructions": {
    "route": "oral",
    "frequency_per_day": 2,
    "duration_days": 90
  },
  "issued_at": "2026-01-12T11:20:00Z"
}
```

Returns `422 Unprocessable Entity` if a severe drug-allergy interaction is detected.

Diagnostic Laboratories

Submit a diagnostic lab order

`POST /labs/orders`

Submits a clinical laboratory requisition (CBC, metabolic panel, lipid panel) to the reference laboratory queue.

Request body:

Field	Type	Required	Description
patient_id	string	Yes	Patient UUID.
ordering_provider_id	string	Yes	Ordering clinician UUID.
panel_code	string	Yes	Laboratory panel code.
priority	string	Yes	Order urgency (`routine` or `stat`).

Response 200 OK:

```
{
  "id": "lab_551902",
  "patient_id": "pat_10293847",
  "panel_code": "CMP-14",
  "status": "queued",
  "observations": [
    {
      "loinc_code": "2345-7",
      "analyte_name": "Glucose [Mass/volume] in Serum or Plasma",
      "value": 94.0,
      "unit": "mg/dL",
      "reference_range": "70-99",
      "abnormal_flag": "normal"
    }
  ],
  "reported_at": "2026-02-01T08:00:00Z"
}
```

Retrieve diagnostic lab results

`GET /v3/labs/orders/{lab_order_id}`

Retrieves quantitative LOINC-coded laboratory observations, reference ranges and abnormal flags for a completed lab requisition. This endpoint is public and does not require authentication.

Path parameters:

Parameter	In	Type	Required	Description
lab_order_id	path	string	Yes	Unique identifier of the laboratory order.

Response 200 OK:

```
{
  "id": "lab_551902",
  "patient_id": "pat_10293847",
  "panel_code": "CMP-14",
```

```
  "status": "final",
  "observations": [
    {
      "loinc_code": "2345-7",
      "analyte_name": "Glucose [Mass/volume] in Serum or Plasma",
      "value": 94.0,
      "unit": "mg/dL",
      "reference_range": "70-99",
      "abnormal_flag": "normal"
    }
  ],
  "reported_at": "2026-02-01T08:00:00Z"
}
```

Insurance Claims

Submit an insurance claim

`POST /claims`

Submits an institutional or professional medical claim (ICD-10 diagnosis codes and CPT procedure lines) to the clearinghouse.

Request body:

```
{
  "patient_id": "pat_10293847",
  "appointment_id": "apt_330192",
  "diagnosis_codes": ["I10", "E11.9"],
  "billed_amount_cents": 28500
}
```

Response 201 Created:

```
{
  "id": "clm_901823",
  "patient_id": "pat_10293847",
  "adjudication_status": "received",
  "billed_amount_cents": 28500,
  "allowed_amount_cents": 21000,
  "patient_copay_cents": 3000,
  "submitted_at": "2026-02-02T17:10:00Z"
}
```

Retrieve insurance claim adjudication status

`GET /claims/{claim_id}`

Path parameters:

Parameter	In	Type	Required	Description
claim_id	path	string	Yes	Unique identifier of the insurance claim.

Response 200 OK:

```
{
  "id": "clm_901823",
  "patient_id": "pat_10293847",
  "adjudication_status": "paid",
  "billed_amount_cents": 28500,
  "allowed_amount_cents": 21000,
  "copay_cents": 3000,
  "submitted_at": "2026-02-02T17:10:00Z"
}
```